Benchmark: **MA.7.NSO.2.2** · Four operations with rational numbers Student code: _____ Calculator: no

Learning goal: I can add, subtract, multiply, and divide rational numbers. (circle your level after scoring)

4	I can solve a multi-step computation with rational numbers and check that the sign and size make sense.
3	I can add, subtract, multiply, and divide fractions, decimals, and negatives accurately.
2	I can add and subtract integers and multiply and divide with the correct sign.
1	I need help to start this benchmark.

1. What is the value of the expression?

$4.13 - 2.91 - 17.5$

2. What is the value of the expression? Write your answer in simplest form.

$-0.64 \div \left(-\frac{4}{5}\right)$

Turn over →

Benchmark: **MA.7.NSO.2.2** · Four operations with rational numbers Student code: _____

3. What number makes the equation true?

$-2\frac{2}{5} + __ = -3.61$

4. Evaluate the expression. Write your answer in simplest form.

$2\frac{1}{2} \times (-0.1)$

5. A student evaluated the expression $-\frac{1}{6} \div 1\frac{4}{5} - \frac{1}{2}$ as shown.

Step 1: $-\frac{1}{6} \times \frac{5}{9} = \frac{5}{54}$
Step 2: $\frac{5}{54} - \frac{1}{2} = -\frac{11}{27}$

In which step did the student make the first error? What is the correct value of the expression?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-NSO.2.2 · MA.7.NSO.2.2 · Four operations with rational numbers

Unit 3: Operations with Rational Numbers. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **−16.28**

Subtract from left to right.

$4.13 - 2.91 = 1.22$.

$1.22 - 17.5 = -16.28$.

2. Regular $\frac{4}{5}$ **(or 0.8)**

Write the decimal as a fraction: $-0.64 = -\frac{16}{25}$.

Dividing by a fraction means multiplying by its reciprocal: $-\frac{16}{25} \times \left(-\frac{5}{4}\right) = \frac{4}{5}$.

Two negatives give a positive.

3. Regular $-1\frac{21}{100} = -\frac{121}{100} (\text{or } -1.21)$

The missing term is the difference of the two known numbers.

Write the fraction as a decimal: $-2\frac{2}{5} = -2.4$.

$-3.61 - (-2.4) = -1.21$, which is $-1\frac{21}{100}$.

Check: $-2\frac{2}{5} + \left(-1\frac{21}{100}\right) = -3.61$.

4. Regular $-\frac{1}{4} (\text{or } -0.25)$

Write the decimal as a fraction: $-0.1 = -\frac{1}{10}$.

Write the mixed number as an improper fraction: $2\frac{1}{2} = \frac{5}{2}$.

$\frac{5}{2} \times \left(-\frac{1}{10}\right) = -\frac{1}{4}$. A negative times a positive is negative.

5. Rigor **Step 1; the correct value is $-\frac{16}{27}$. Score: 1 point for the step and the correct value; the step alone earns 0.**

Step 1 has the wrong sign: a negative divided by a positive (or a positive by a negative) is negative, and two negatives give a positive.

Correct work: $-\frac{1}{6} \times \frac{5}{9} = -\frac{5}{54}$; $-\frac{5}{54} - \frac{1}{2} = -\frac{16}{27}$.

Re-check strip (3 items, C-NSO.2.2-R)

1. Regular **−45.75**

Subtract from left to right.

$6.49 - 2.34 = 4.15$.

$4.15 - 49.9 = -45.75$.

2. Regular **−6**

Write the decimal as a fraction: $-1.5 = -\frac{3}{2}$.

Dividing by a fraction means multiplying by its reciprocal: $-\frac{3}{2} \times \frac{4}{1} = -\frac{12}{2} = -6$.

A negative divided by a positive (or a positive by a negative) is negative.

3. Regular $\frac{5}{6}$

LCD of 3 and 6 is 6.

$\frac{4}{6} - \left(-\frac{1}{6}\right) = \frac{5}{6}$.

Comparison strip (2 items, C-NSO.2.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $-\frac{3}{4}$

LCD of 2 and 4 is 4.

$-\frac{2}{4} - \frac{1}{4} = -\frac{3}{4}$.

2. Regular $-\frac{8}{15} (\text{or } -0.5\overline{3})$

Write the decimal as a fraction: $-1.6 = -\frac{8}{5}$.

$\frac{1}{3} \times \left(-\frac{8}{5}\right) = -\frac{8}{15}$. A negative times a positive is negative.

Reteach plan — C-NSO.2.2

MA.7.NSO.2.2 · Four operations with rational numbers

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-NSO.2.2-R

Benchmark: **MA.7.NSO.2.2** · Four operations with rational numbers Student code: _____ Calculator: no

1. What is the value of the expression?

$6.49 - 2.34 - 49.9$

2. What is the value of the expression? Write your answer in simplest form.

$-1.5 \div \frac{1}{4}$

3. Subtract. Write your answer in simplest form.

$\frac{2}{3} - \left(-\frac{1}{6}\right)$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-NSO.2.2-R

Benchmark: **MA.7.NSO.2.2** · Four operations with rational numbers Student code: _____ Calculator: no

1. What is the value of the expression?

$6.49 - 2.34 - 49.9$

2. What is the value of the expression? Write your answer in simplest form.

$-1.5 \div \frac{1}{4}$

3. Subtract. Write your answer in simplest form.

$\frac{2}{3} - \left(-\frac{1}{6}\right)$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-NSO.2.2-X

Benchmark: **MA.7.NSO.2.2** · Four operations with rational numbers

Student code: _____ Calculator: no

1. Evaluate. Write your answer in simplest form.

$-\frac{1}{2} - \frac{1}{4}$

2. Evaluate the expression. Write your answer in simplest form.

$\frac{1}{3} \times (-1.6)$

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-NSO.2.2-X

Benchmark: **MA.7.NSO.2.2** · Four operations with rational numbers

Student code: _____ Calculator: no

1. Evaluate. Write your answer in simplest form.

$-\frac{1}{2} - \frac{1}{4}$

2. Evaluate the expression. Write your answer in simplest form.

$\frac{1}{3} \times (-1.6)$

Score: ____ / 2 → **copy to your progress strip**